

Intro to Basis Sets

What do the letters even mean?

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Schrödinger Equation

$$\hat{H}\Psi = E\Psi$$

Problem

We can't determine the exact wavefunction for many-electron systems.

Objective

Construct wavefunctions using molecular orbitals as approximate solutions to the Schrödinger equation.

We construct wavefunctions using basis sets.

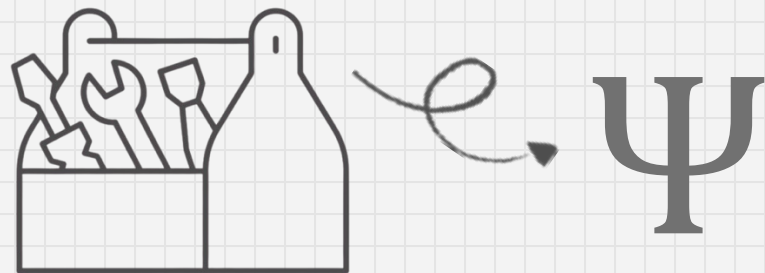
Objectives

1. What are basis sets?
2. Types of basis functions
3. Basis set classification
4. Basis set anatomy
5. Basis set families

What is a basis set?

What is a basis set?

In quantum chemistry, a basis set is a set of functions (called basis functions) that are linearly combined to create molecular orbitals.



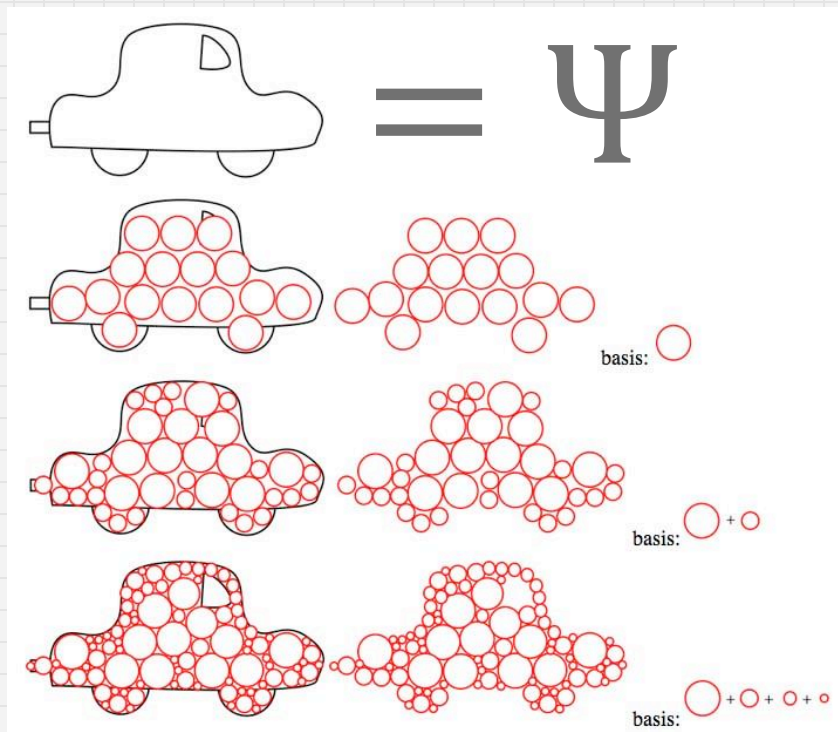
What is a basis set?

In quantum chemistry, a basis set is a set of functions (called basis functions) that are linearly combined to create molecular orbitals.

$$\underbrace{\phi_i(\mathbf{r})}_{\text{constructed molecular orbital}} = \sum_j \overset{\text{MO mixing coefficient}}{\underbrace{c_{ij}}_{\uparrow}} \underbrace{\chi_j(\mathbf{r})}_{\text{basis function}}$$

These functions are usually atomic orbitals centered on atoms, but can theoretically be any function.

What is a basis set?



Increasing the number of basis functions gives a more detailed representation of the wavefunction.

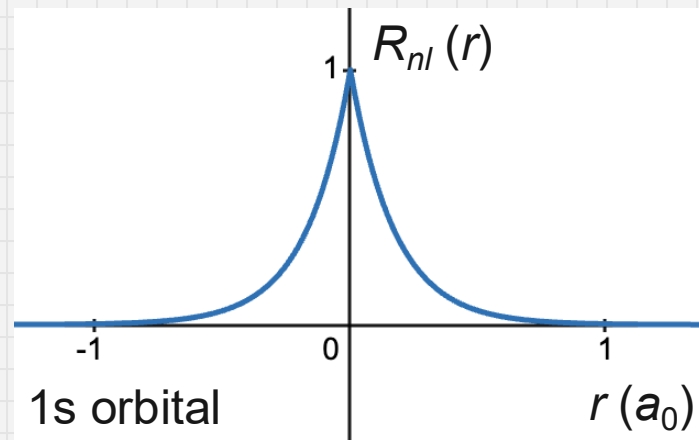
Basis sets can come in hierarchies of increasing size and be systematically extended to the complete basis set (CBS) limit.

Types of basis functions

1. Slater-Type Orbitals (STOs)

The exact solutions to the hydrogenic wavefunction take the form of Slater functions.

Slater-Type Orbitals (STOs) are constructed to resemble these Slater functions.



$$\phi_{abc}^{STO}(x, y, z) = Nx^a y^b z^c e^{-\zeta r}$$

1. Slater-Type Orbitals (STOs)

Pros

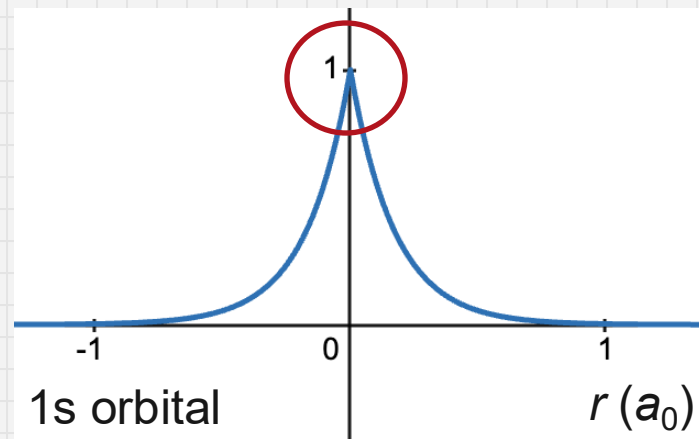
A physically accurate choice:

- Cusp at the nucleus
- Exponential decay away from the nucleus

Cons

Cannot do two-electron integrals analytically

STOs are accurate but are far more computationally difficult

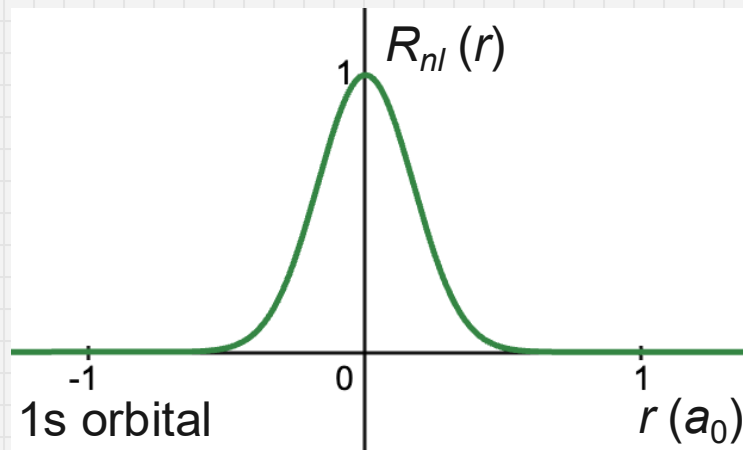


$$\phi_{abc}^{STO}(x, y, z) = Nx^a y^b z^c e^{-\zeta r}$$

2. Gaussian-Type Orbitals (GTOs)

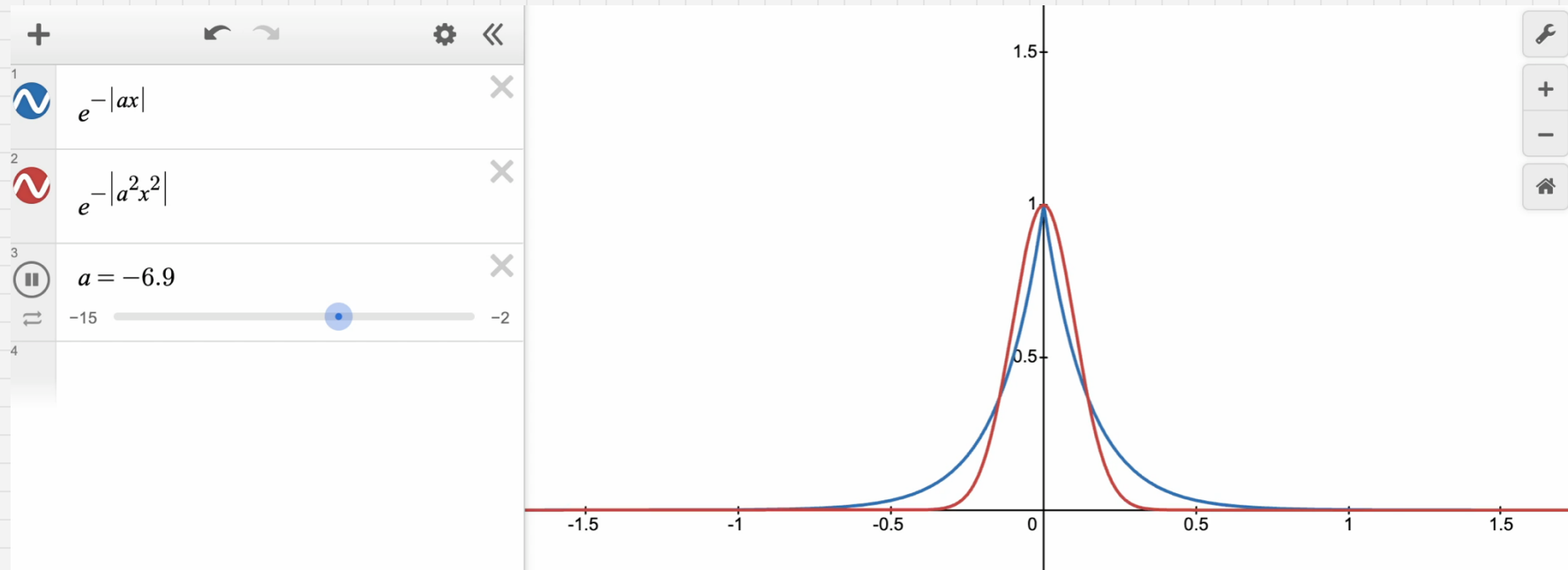
Called **primitive Gaussians**

- Computationally tractable due to Gaussian product theorem
- Exponential decay away from the nucleus, albeit too rapid
- Wrong cusp behavior
- Less accurate, but what if we combine GTOs?



$$\phi_{abc}^{GTO}(x, y, z) = Nx^a y^b z^c e^{-\zeta r^2}$$

STO and GTO Widths



3. Contracted Gaussian-Type Orbitals

CGTOs: linear combination of primitive Gaussians to represent our Slater-type orbitals

Pre-combine primitive Gaussians in fixed proportions

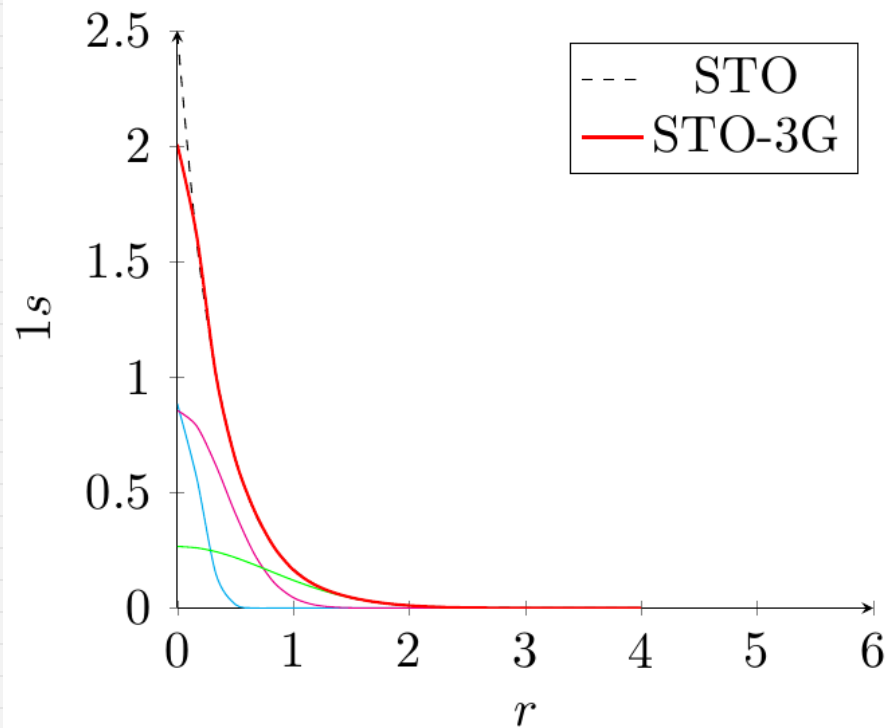
Contracted Gaussians act like a single basis function even though they contain multiple primitive Gaussians underneath.

3. Contracted Gaussian-Type Orbitals

Example

STO-nG

n = number of
primitive Gaussians

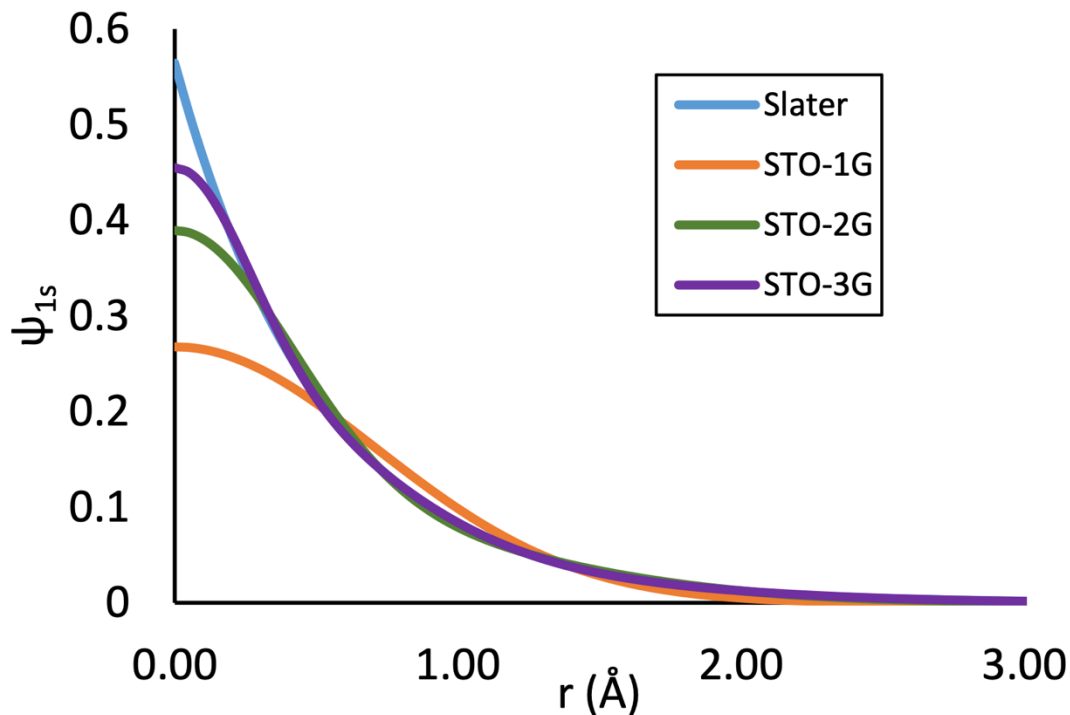


3. Contracted Gaussian-Type Orbitals

Example

STO-nG

n = number of
primitive Gaussians



Basis Set Classification

How many and what types of functions are included in basis sets?

Basis Set Classification

Minimal: One basis function (STO, GTO, or CGTO) for each atomic orbital (AO) in the atom

Double-zeta (DZ): Two basis function for each AO

Triple-zeta (TZ): Three basis function for each AO

QZ, 5Z, 6Z, ... etc.

Split-valence basis uses only **one** basis function for each core AO and a **larger** basis for the valence AOs.

Basis Set Classification

Minimal

DZ

TZ

QZ

5Z

6Z

... etc.



Adding more basis functions.

Provides a larger menu of differently-sized basis functions to grab when building molecular orbitals.

Better representation of the system (usually).

Examples

How many basis functions are needed?

1. H atom, minimal basis

H: $1s^1 \rightarrow 1 \text{ AO} \times 1 \text{ basis function per AO} = 1 \text{ basis function}$

2. C atom, minimal basis

C: $1s^2 2s^2 \underline{2p^2} \rightarrow 5 \text{ AOs} \times 1 \text{ basis function per AO} = 5 \text{ basis functions}$
 $2p_x, 2p_y, 2p_z$

Examples

How many basis functions are needed?

3. C atom, double-zeta basis

C: $1s^2 2s^2 2p^2 \rightarrow 5 \text{ AOs} \times 2 \text{ basis functions per AO} = 10 \text{ basis functions}$

4. C atom, split-valence double-zeta basis

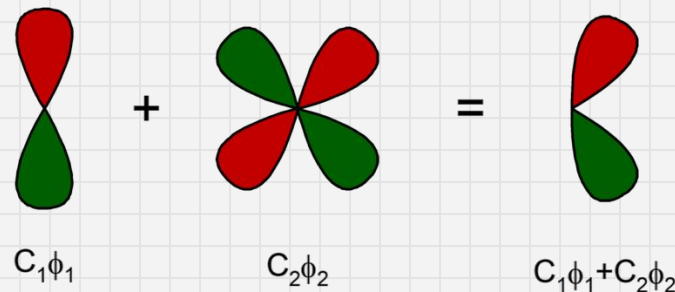
C: $1s^2 2s^2 2p^2 \rightarrow 1 \text{ core AOs} \times 1 \text{ basis function per AO} = 1$
 $+ 4 \text{ valence AOs} \times 2 \text{ basis functions per AO} = 8$

9 basis functions

Polarization Functions

An atom's orbitals might want to distort if they are approached by other atoms.

General: polarizing a basis function with l angular momentum requires mixing in basis functions with $l+1$ angular momentum



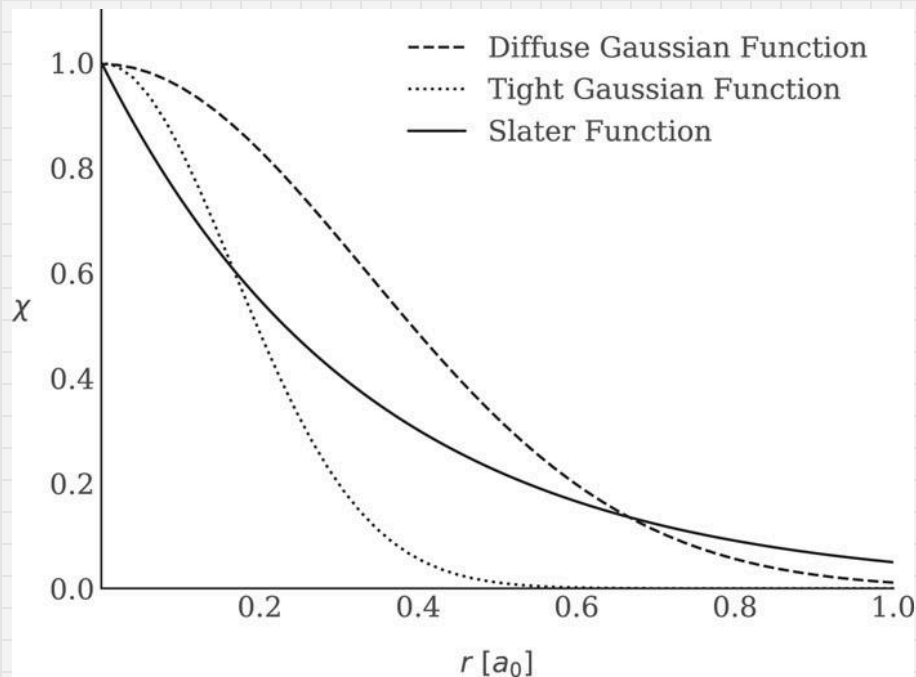
Polarizing an s-orbital requires a p-orbital.

Polarizing an p-orbital requires a d-orbital.

Diffuse Functions

Small ζ exponents $\rightarrow$
electron is further from
nucleus

Needed for anions and
compounds with lone pairs,
etc.



Basis Set Anatomy

credit to Dr. Sherrill for this heading!

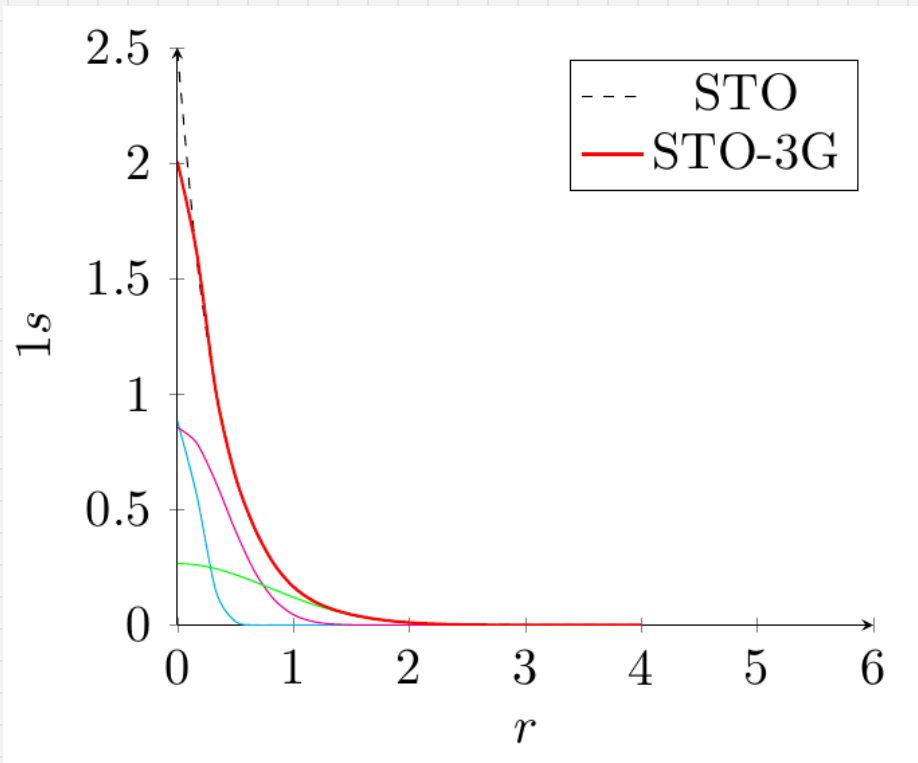
Basis Set Anatomy

STO-3G

CGTO with 3 primitive
gaussians

Need to specify

- ζ_i coefficients for each Gaussian
- c_i weighting coefficients



Basis Set Anatomy

STO-3G, Psi4 Format

atom symbol **H**

S

one basis function

0 **3** $\rightarrow$ # primitive gaussians

1.00

0.3425250914D+01 0.1543289673D+00

0.6239137298D+00 0.5353281423D+00

0.1688554040D+00 0.4446345422D+00

ζ_i coefficients c_i weighting coefficients

Basis Set Anatomy

STO-3G, Psi4 Format

	C	0			
1s orbital basis function	S	3	1.00		
			0.7161683735D+02	0.1543289673D+00	
			0.1304509632D+02	0.5353281423D+00	
			0.3530512160D+01	0.4446345422D+00	
combined SP shell (quirky syntax)	SP	3	1.00		
			0.2941249355D+01	-0.9996722919D-01	0.1559162750D+00
			0.6834830964D+00	0.3995128261D+00	0.6076837186D+00
			0.2222899159D+00	0.7001154689D+00	0.3919573931D+00
	****	ζ_i coefficients	2s c_i weighting coefficients	2p c_i weighting coefficients	

Basis Set Exchange (BSE)

x

+

←

→

↺

basissetexchange.org

☆

📅

📁

🔍

🔴

New Chrome available

⋮

🗄️

📧 Vandy Email

📧 UGA Mail

🗨️ Discord | CCQC

📅 Google Calendar

🔑 Athena Login

📄 ELC

📌 Google Keep

🍴 Recipes

📖 All Bookmarks



Basis Set Exchange

DownloadGitHubFeedbackAboutHelp▼Request a Basis set

All roles▼All▼

2ZaP

2ZaPa-NR

2ZaPa-NR-CV

3-21G

3ZaP

3ZaPa-NR

3ZaPa-NR-CV

4-31G

4ZaP

4ZaPa-NR

4ZaPa-NR-CV

5-21G

5ZaP

5ZaPa-NR

5ZaPa-NR-CV

6-21G

6-31++G

6-31++G*

6-31++G**

6-31++G**~J

6-31+G

6-31+G*

6-31+G*~J

6-31+G**

6-31+G**~J

search basis sets...

References for selected basis

Plain Text

Get References

Total found: 748 basis sets

Select All

Reset Selection

1 H																	2 He				
3 Li	4 Be															5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg															13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr				
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe				
55 Cs	56 Ba			72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn			
87 Fr	88 Ra			104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og			
57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu							
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr							

Download basis set

Format

Psi4

⌵

Get Basis Set

Advanced

<https://github.com/MolSSI-BSE>

Basis Sets Families

```
graph TD; A[Basis Sets Families] --> B[Pople]; A --> C[Dunning]; A --> D[Ahlrichs/Karlsruhe];
```

Pople

Dunning

**Ahlrichs/
Karlsruhe**

Pople Basis Sets

Pople Basis Sets



John Pople,
Nobel Laureate

Minimal: STO-3G

Split-valence

1 basis function for core AOs;
larger basis for valence AOs

Pople Basis Sets



John Pople,
Nobel Laureate

Minimal: STO-3G

Split-valence
Double-zeta

core valence

X - **YZ** **G**

X = # of primitive
Gaussians comprising each
core AO basis function

Y = # of primitive Gaussians
in 1st basis function
Z = # of primitive Gaussians
in 2nd basis function

Pople Basis Sets



John Pople,
Nobel Laureate

Minimal: STO-3G

Split-valence
Double-zeta

Core orbital is a CGTO
made of 6 primitive
Gaussians

core valence
6-31G

Valence is described by 2
orbitals:

- 1 CGTO with 3 primitive Gaussians
- 1 single Gaussian (GTO)

Pople Basis Sets



John Pople,
Nobel Laureate

Minimal: STO-nG

Split-valence

Double-zeta: 6-31G, 3-21G

Triple-zeta: 6-311G

Pople Basis Sets



John Pople,
Nobel Laureate

Polarization *

$$6-31G^* = 6-31G(d)$$

6-31G with d polarization functions
on heavy atoms

$$6-31G^{**} = 6-31G(d,p)$$

6-31G* augmented with p
polarization functions for hydrogen

Still DZ!

Pople Basis Sets



John Pople,
Nobel Laureate

Diffuse Functions +

6-31+G

6-31G with s and p diffuse functions
for heavy atoms

6-31++G

6-31+G augmented with s diffuse
functions on hydrogen

Still DZ!

Pople Basis Sets



John Pople,
Nobel Laureate

Putting it all together...

6-311G(3df,2pd)

- 6-311G base $\rightarrow$ split-valence, triple-zeta
- Core: 1 CGTO with 6 primitive Gaussians
- Valence: 1 CGTO with 3 primitive Gaussians, 2 single, primitive Gaussians (GTOs)
- Add 3 sets of d functions and 1 set of f functions to heavy atoms
- Add 2 sets of p functions and 1 set of d functions to hydrogens

Dunning Basis Sets

Dunning Basis Sets



Thom Dunning Jr.

Basis sets optimized for Hartree-Fock computations are not ideal for correlated methods.

Developed “correlation consistent” basis sets designed to converge to the complete basis set (CBS) limit.

cc-pV X Z, X = D, T, Q, 5, 6, ...

correlation-consistent (cc)
polarized valence (pV)
 X -zeta basis

Dunning Basis Sets



Thom Dunning Jr.

Example

C atom, cc-pVDZ

C: $1s^2 2s^2 2p^2$

Split-valence, double-zeta:

- 1 s function for core

- 1 x 2 = 2 s functions for valence

- 1 x 2 = 2 p functions for valence

Polarized: add 1 set of d functions

Set of functions: 3s 2p 1d

Dunning Basis Sets



Thom Dunning Jr.

Example

C atom, cc-pVTZ

C: $1s^2 2s^2 2p^2$

Split-valence, triple-zeta:

- 1 s function for core

- $1 \times 3 = 3$ s functions for valence

- $1 \times 3 = 3$ p functions for valence

Polarized: add 2 sets of d functions, 1 set of f functions

Set of functions: 4s 3p 2d 1f

Dunning Basis Sets



Thom Dunning Jr.

cc-pV XZ vs. cc-pC VXZ

Adds functions near the nucleus to describe core electron correlation.

Use for non-frozen core computations.

cc-pV($X+d$)Z

Use for Aluminum-Argon
cc-pVXZ d-polarization functions were not tight enough

Dunning Basis Sets



Thom Dunning Jr.

What about diffuse functions?

Augmented basis set (aug): add one set of diffuse functions for every type of angular momentum present (e.g. s, p, d, f, etc.)

C atom, cc-pVDZ: 3s 2p 1d

aug-cc-pVDZ: add diffuse s, p, d

Truhlar's Calendar Notation



Cheaper versions of aug-cc-pVXZ by removing some diffuse functions.

AUG One set of diffuse functions for every / type.

JUL Removes diffuse functions on hydrogen.

JUN Removes diffuse functions on hydrogen and also the next highest orbital for heavy atoms.

MAY Removes next-highest...

...

Ahrichs/Karlsruhe

Basis Sets

Ahlrichs/Karlsruhe Basis Sets



Reinhart
Ahlrichs

Split-valence basis sets. Previous variations just “def”

def2-

double-	SV(P)
zeta	SVP
	SVPPD
triple-	TZVP
zeta	TZVPD
	TZVPPD
quad-	QZVP
ruple-	QZVPD
zeta	QZVPP
	QCVPPD

P: polarized

PP: (usually) more polarized

D: add more diffuse subshells

Less costly than Dunning “aug”

SVP ~ cc-pVDZ; TZVP ~ cc-pVTZ

Often recommended basis set family for
DFT computations.

Conclusions

1. A basis set is a set of basis functions combined in linear combinations to create molecular orbitals.
2. STOs, GTOs, and CGTOs are types of basis functions.
3. Basis sets can be classified by how many basis functions represent each atomic orbital.
4. The three main families of basis sets used in computational chemistry: Pople, Dunning, and Ahlrichs/Karlsruhe families.

Primary reference/resource: Dr. Sherrill's basis set lectures on YouTube

Questions?